

Automated Finish Classification for Veridian Lacquer Trend Reports

Veridian Lacquer is a boutique nail polish manufacturer seeking to standardize the classification of nail polish finishes for their annual state of the shade report. Traditionally, finishes such as Creme, Shimmer, Metallic, and Holographic are classified manually, which introduces inconsistency.

The goal of this project is to develop a machine learning model in MATLAB that automatically classifies nail polish finishes based on measurable color properties: RGB values (Red, Green, Blue), Opacity (0-100), Reflectance Index (0-1). A synthetic dataset is generated to simulate realistic nail polish finish characteristics across four categories: Creme, Shimmer, Metallic, and Holographic. Each class was defined using controlled ranges of RGB values, opacity levels, and reflectance indices to reflect typical visual properties. To ensure consistency and model performance, all numerical features were normalized using standard score normalization, which scales the data to have a mean of zero and a standard deviation of one. This step is particularly important for distance based algorithm such as k-nearest neighbours (k-NN), ensuring that no single feature disproportionately influence the classification. On Exploratory data analysis, the scatter plots reveal distinct clustering patterns among the four finish categories. Metallic and Holographic finishes tend to exhibit higher reflective values and on the observation in this feature space, they are clearly separable. Creme and Shimmer finishes exhibit some overlap when considering only Red and Reflectance. This indicates that additional features such as Green, Blue, Opacity are necessary to fully distinguish between these classes. And on the 3D plot since both Metallic and Holographic have high reflectance, overlapping opacity, and RGB differences are not strong enough on the given reflectance range. Reflectance ranges were adjusted to improve separability between Metallic and Holographic finishes. A slight overlap was intentionally retained to simulate realistic conditions where similar finishes may share visual properties, ensuring the classification model relies on multiple features rather than a single threshold.

The dataset is split into training and testing sets to evaluate how well the model generalizes to unseen data. An 80/20 split is used for this project as is commonly used to balance training performance and evaluation reliability. A k-NN classification model was trained using the normalized feature set. The model was then evaluated on unseen test data to assess its generalization performance. The confusion matrix reveals strong classification performance, with small number of misclassification. Specifically, a few Shimmer samples were incorrectly classified as Creme and Metallic, while some Metallic and Holographic samples were confused with each other. Despite these minor overlaps, the overall model accuracy reached 95%, indicating that the selected features and classification approach are effective for this task. In addition to the k-NN model, a Decision Tree classifier was implemented to compare different approaches to the classification task. While k-NN is a distance based algorithm that classifies samples based on similarity in feature space, the Decision Tree model uses a series of rule based splits to partition the data. This comparison allows for evaluating how different modelling techniques perform on the same dataset and provides insight into the most effective method for distinguishing between nail polish finishes.

Synthetic Dataset Generation

```
clc; clear; close all;  
% Creating synthetic data  
rng(1); % ensure same random data every time  
  
% generate synthetic dataset  
n = 150;
```

```

%***** Creme Finish *****%

% High opacity, low reflectance, smooth color distribution

R1 = randi([100 255],n,1);
G1= randi([100 255],n,1);
B1 = randi([100 255],n,1);
Opacity1 = randi([80 100],n,1);
Reflect1 = rand(n,1) * 0.3;

% *****Shimmer Finish *****
% Medium opacity, moderate reflectance, more variation
R2 = randi([80 255],n,1);
G2 = randi([80 255],n,1);
B2 = randi([80 255],n,1);
Opacity2 = randi([50 80],n,1);
Reflect2 = 0.3 + rand(n,1) * 0.3;

% ***** Metallic Finish *****
% High reflectance, moderate opacity
R3 = randi([50 200],n,1);
G3 = randi([50 200],n,1);
B3 = randi([50 200],n,1);
Opacity3 = randi([60 90],n,1);
Reflect3 = 0.6 + rand(n,1) * 0.25;

% ***** Holographic Finish *****
% High reflectance and high RGB variability
R4 = randi([0 255],n,1);
G4 = randi([0 255],n,1);
B4 = randi([0 255],n,1);
Opacity4 = randi([40 80],n,1);
Reflect4 = 0.75 + rand(n,1) * 0.25;

% combine all classes into one dataset
R = [R1; R2; R3; R4];
G = [G1; G2; G3; G4];
B = [B1; B2; B3; B4];
Opacity = [Opacity1; Opacity2; Opacity3; Opacity4];
Reflectance = [Reflect1; Reflect2; Reflect3; Reflect4];

% Create labels

Label = [repmat("Creme", n, 1);
         repmat("Shimmer", n, 1);
         repmat("Metallic", n, 1);
         repmat("Holographic", n,1)];

% Create table

```

```
data = table(R,G,B, Opacity, Reflectance, Label);
% display first few rows
disp('Sample of generated dataset');
```

Sample of generated dataset

```
tail(data,20);
```

R	G	B	Opacity	Reflectance	Label
5	176	224	63	0.90266	"Holographic"
252	14	20	50	0.80323	"Holographic"
156	41	47	52	0.81244	"Holographic"
96	183	180	47	0.77804	"Holographic"
6	248	235	47	0.8269	"Holographic"
99	237	64	59	0.87436	"Holographic"
115	230	5	60	0.87245	"Holographic"
238	32	245	60	0.99889	"Holographic"
71	37	174	80	0.99528	"Holographic"
157	18	79	45	0.85914	"Holographic"
170	138	120	71	0.99301	"Holographic"
47	225	153	72	0.8036	"Holographic"
129	136	95	63	0.88528	"Holographic"
201	200	223	58	0.86336	"Holographic"
24	48	185	76	0.9919	"Holographic"
155	220	244	73	0.95904	"Holographic"
6	47	89	42	0.80354	"Holographic"
1	14	239	79	0.85988	"Holographic"
15	107	122	40	0.79212	"Holographic"
123	142	164	63	0.88227	"Holographic"

```
summary(data);
```

data: 600x6 table

Variables:

```
R: double
G: double
B: double
Opacity: double
Reflectance: double
Label: string
```

Statistics for applicable variables:

	NumMissing	Min	Median	Max	Mean	Std
R	0	1	151	255	149.4583	58.4192
G	0	3	154	255	149.4850	60.0799
B	0	0	152	255	149.7150	59.4060
Opacity	0	40	72	100	72.2183	14.7113
Reflectance	0	9.3386e-04	0.6001	0.9989	0.5494	0.2857
Label	0					

Feature Engineering

```
% Feature normalization
% extract numerical features
X = data{:,1:5};
```

```
% Normalize data (mean = 0, std = 1)
X_norm = normalize(X);

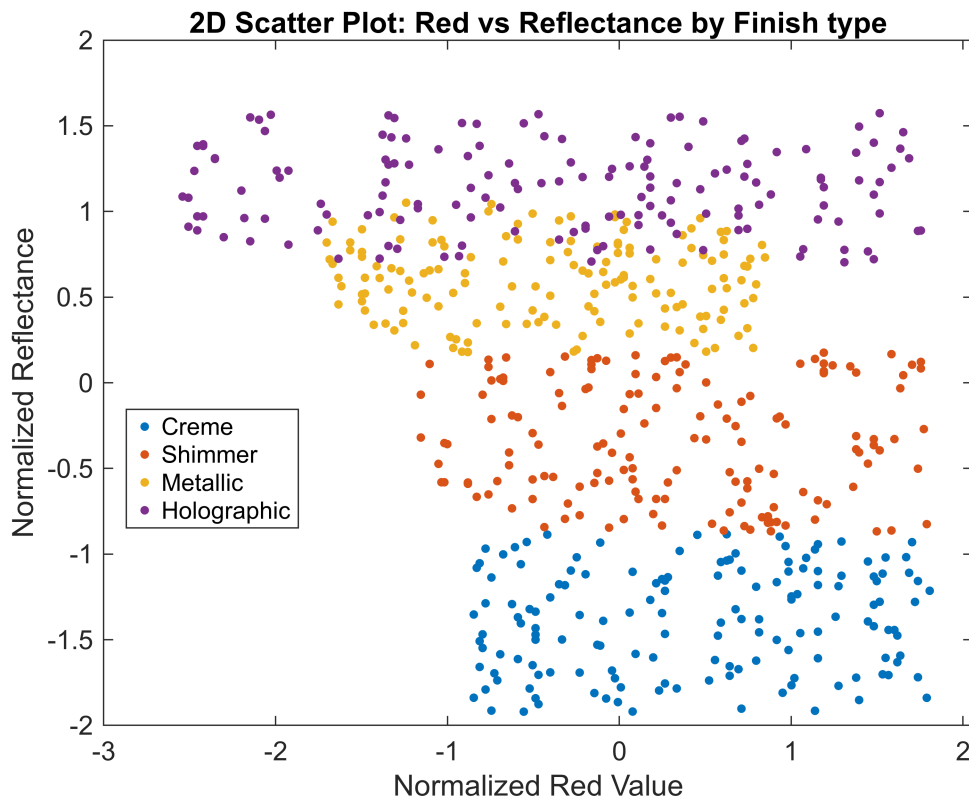
% extract labels
Y = data.Label;

disp('Feature normalization complete.');
```

Feature normalization complete.

Exploratory data analysis

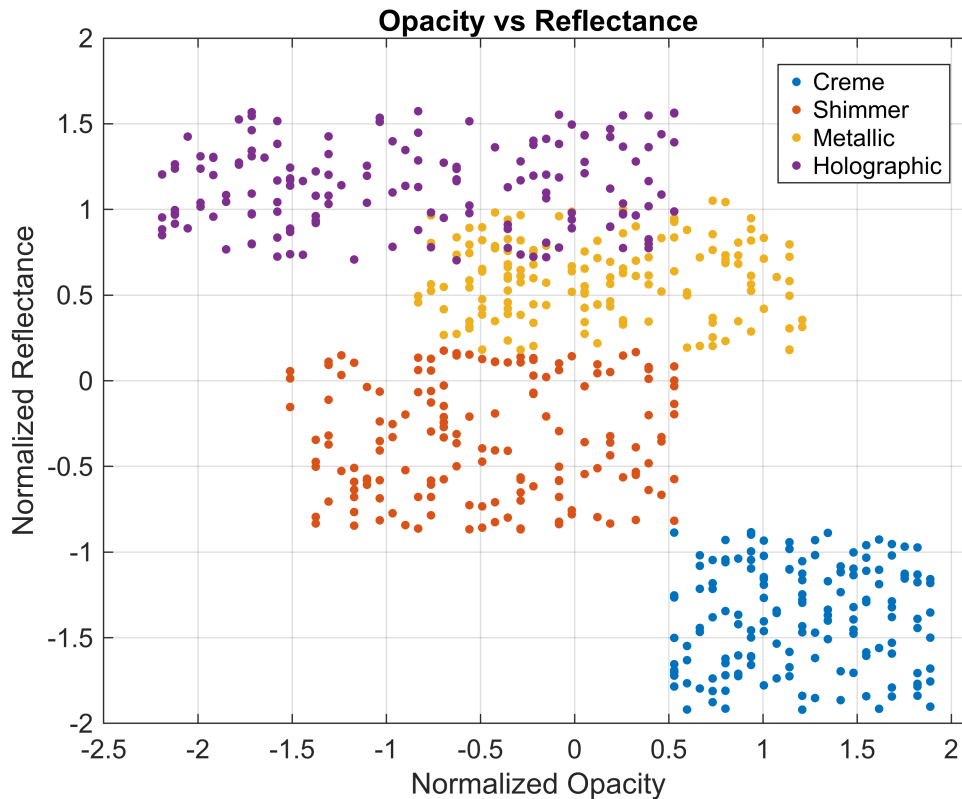
```
% Visualize relationships between features and class labels
figure
gscatter(X_norm(:,1), X_norm(:,5), Y);
xlabel('Normalized Red Value');
ylabel('Normalized Reflectance');
title('2D Scatter Plot: Red vs Reflectance by Finish type');
```



Opacity vs Reflectance

Creme and Shimmer clearly separated, while Metallic and Holographic somehow mixed. So, from feature to feature the separation performance is different.

```
figure
gscatter(X_norm(:,4), X_norm(:,5), Y)
xlabel('Normalized Opacity');
ylabel('Normalized Reflectance');
title('Opacity vs Reflectance');
grid on;
```



3D Scatter Plot

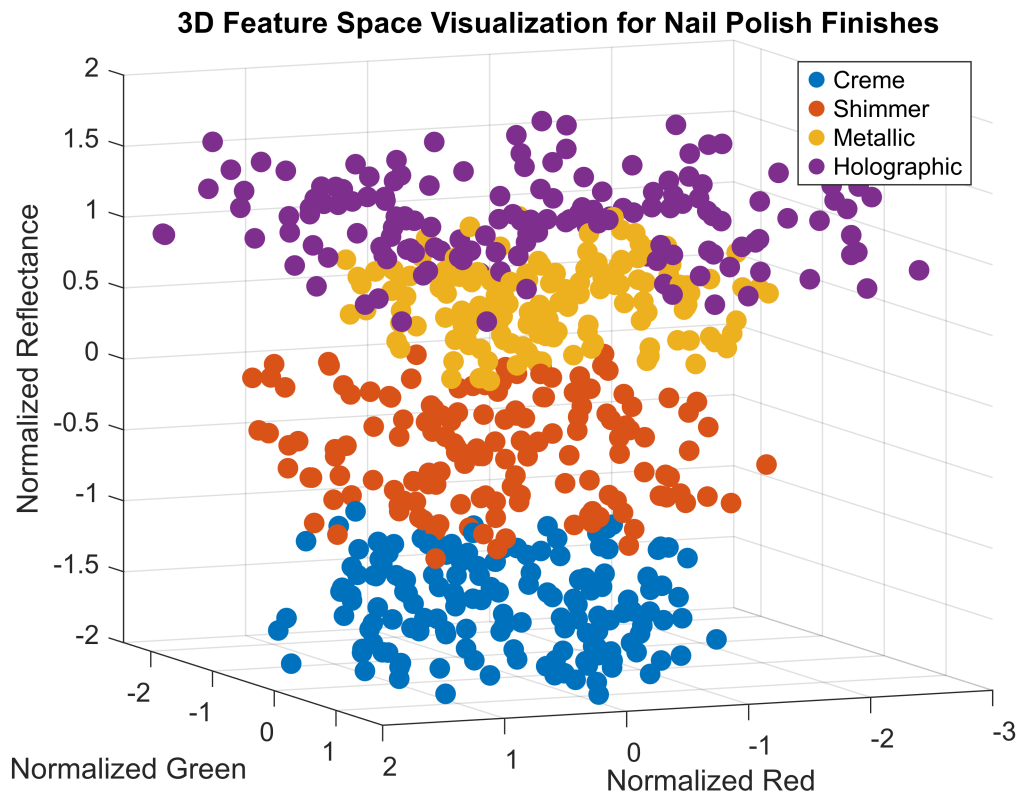
```
% 3D plot visualization
figure
hold on

% logical indexing for each class
idx1 = Y == "Creme";
idx2 = Y == "Shimmer";
idx3 = Y == "Metallic";
idx4 = Y == "Holographic";

% Plot each class separately
scatter3(X_norm(idx1,1),X_norm(idx1,2),X_norm(idx1,5), 60, 'filled');
scatter3(X_norm(idx2,1),X_norm(idx2,2),X_norm(idx2,5), 60, 'filled');
scatter3(X_norm(idx3,1),X_norm(idx3,2),X_norm(idx3,5), 60, 'filled');
scatter3(X_norm(idx4,1),X_norm(idx4,2),X_norm(idx4,5), 60, 'filled');
```

```
%Labels
```

```
xlabel('Normalized Red');  
ylabel('Normalized Green');  
zlabel('Normalized Reflectance');  
title('3D Feature Space Visualization for Nail Polish Finishes');  
legend('Creme', 'Shimmer', 'Metallic', 'Holographic');  
grid on;  
view(157,9)  
exportgraphics(gcf, '3D_Finish_Classification.png', 'Resolution', 300);
```



Train /Test Split

```
% Split data into Training and Testing Sets  
% We split the dataset into 80% training and 20% testing  
cv = cvpartition(Y, 'HoldOut', 0.2);  
% Training Data  
Xtrain = X_norm(training(cv), :);  
Ytrain = Y(training(cv));  
  
% Testing data  
Xtest = X_norm(test(cv), :);  
Ytest = Y(test(cv));  
  
disp('Data split complete:')
```

Data split complete:

```
fprintf('Training samples: %d\n', length(Ytrain));
```

Training samples: 480

```
fprintf('Testing samples: %d\n', length(Ytest));
```

Testing samples: 120

Model Training

```
% Train k-Nearest Neighbors(k-NN) Model
% k = 5 neighbors
model = fitcknn(Xtrain, Ytrain, 'NumNeighbors', 5);
disp('Model training complete');
```

Model training complete

```
% Train Decision Tree classifier
treeModel = fitctree(Xtrain, Ytrain);
disp('Decision Tree model training complete');
```

Decision Tree model training complete

Prediction

```
% Predict on test data
Ypred = predict(model, Xtest);
disp('Prediction complete')
```

Prediction complete

```
%predict using decision tree
Ypred_tree = predict(treeModel, Xtest);
```

Confusion Matrix

```
% Model Evaluation using Confusion Matrix
%convert into categorical
Ytest = categorical(Ytest);
Ypred = categorical(Ypred);

figure
confusionchart(Ytest, Ypred)
title('Confusion Matrix for Nail Polish Finish Classification');
```

True Class

Creme	30			
Holographic		27	3	
Metallic		1	29	
Shimmer	1		1	28
	Creme	Holographic	Metallic	Shimmer

```
% confusion matrix decision tree
Ytest_cat = categorical(Ytest);
Ypred_tree_cat = categorical(Ypred_tree);

figure
confusionchart(Ytest_cat, Ypred_tree_cat);
title('Decision Tree Confusion Matrix for Nail Polish Finish Classification');
exportgraphics(gcf, 'decision_tree_confusion_matrix.png', 'Resolution', 300);
```

True Class

Creme	30			
Holographic		28	2	
Metallic		1	29	
Shimmer				30
	Creme	Holographic	Metallic	Shimmer

Accuracy Calculation

```
% calculate model Accuracy
accuracy = sum(Ypred == Ytest) / numel(Ytest);
fprintf('Model Accuracy: %.2f%%\n', accuracy * 100);
```

Model Accuracy: 95.00%

```
treeAccuracy = sum(Ypred_tree == Ytest) / numel(Ytest);  
fprintf('Decision Tree Accuracy: %.2f%%\n', treeAccuracy * 100);
```

Decision Tree Accuracy: 97.50%

Saving Model

```
save('finish_model.mat', 'model');
save('decision_tree_model.mat', 'treeModel');
```

Conclusion

This project successfully developed an automated classification system for nail polish finishes using a k-Nearest Neighbors(k-NN) and Decision tree models in MATLAB.

A Synthetic dataset was generated to simulate realistic polish properties, including RGB values, opacity and reflectance, across four finish categories: Creme, Shimmer, Metallic, and Holographic.

The trained k-NN model achieved an overall classification accuracy of approximately 95% on the test dataset. The confusion matrix shows strong performance, with high values along the diagonal indicating correct predictions for most samples. Minor misclassifications were observed, particularly between Shimmer and Creme, and between Metallic and Holographic finishes. overlapping feature characteristics, especially in reflectance and opacity which can make certain finishes visually similar. And the Decision Tree classifier achieved a higher accuracy of approximately 97.5%. The confusion matrix shows near-perfect classification across all four finish categories, with only minor misclassifications observed between Metallic and Holographic finishes. These errors are expected due to the similarity in reflectance and color variability between these two classes, which makes them harder to distinguish using simple threshold based splits.

Over all, both models demonstrate strong capability in distinguishing between nail polish finishes using multiple features. However, Decision Tree model slightly outperforms the k-NN model due to its ability to capture nonlinear decision boundaries effectively.

Future improvements could include incorporating additional features, such as texture or spectral data, and testing more advanced models to further enhance classification accuracy.